Yanbo Zhou

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About me

I have extensive experience across both industry and academia. My work spans memory and storage systems for cloud & AI infrastructure, with a focus on resource efficiency, performance, and reliability at large-scale systems.

Education

University of California, San Diego, La Jolla, USA
Ph.D. in Computer Science — Sep. 2023 - Jun. 2027 (expected)
Advisor: Prof. Steven Swanson

East China Normal University, Shanghai, China
M.Eng. in Software Engineering — Sep. 2016 - Jan. 2019

Taiyuan University of Technology, Taiyuan, China
B.Eng. in Software Engineering — Sep. 2012 - Jun. 2016

Experience

Samsung Semiconductor, Inc., San Jose, USA
Research Intern, Global Open-ecoSystem Team (GOST) — Jun. 2024 - Present

Research intern at Samsung GOST during the summers of 2024, 2025, and 2026. I led following research collaboration projects between UCSD Non-Volatile Systems Lab and Samsung GOST.

- **Memory Pooling for LLM KV Cache**: a CXL-based pooling for multi-tenant KV Cache sharing (ongoing).
- **Software-defined Elastic Memory**: a software-defined elastic memory system for virtual machines, which achieves flexible allocation of memory capacity and bandwidth and performance isolation (published at OSDI'26).
- **Energy-efficient Storage System**: a fast and sustainable storage system for all-flash servers with a CPU scheduler for storage software stack, which enables high performance and energy efficiency (published at SOSP'25).

Alibaba Cloud, Hangzhou, China
Senior Software Engineer, Elastic Block Storage (EBS) Team — Jan. 2019 - Jun. 2023

Worked on block storage products, including both local disks and EBS, as listed below. The local disk products I led was awarded Best Paper at USENIX FAST'26.

- **I2/I2ne Instance**: the second-gen high-performance and low-latency local disk. This is a software-optimized architecture towards NVMe TLC SSDs [product].
- **D3c Instance**: the third-gen cost-effective and large-capacity local disk. This is the first time enabling high-density NVMe QLC SSDs and Zoned Namespace (ZNS) technology in Alibaba Cloud [product].
- **I4 Instance**: the forth-gen high-performance and low-latency local disk. This is a software/hardware co-designed architecture (an ASIC/SoC co-designed DPU) optimized for NVMe TLC SSDs and bare-metal [product].
- **ESSD-PLX**: the fastest elastic block storage volume of Alibaba Cloud based on Optane Persistent Memory cache and RDMA technology [product].

Intel Corporation, Shanghai, China
Software Engineer Intern, Storage Performance Development Kit (SPDK) Team — Jul. 2017 - Dec. 2018

Worked on SPDK project and contributed to the following project:

- **Vhost-NVMe**: First full-stack userspace NVMe virtualization solution. This technology is used as standard storage virtualization protocol for multiple storage products in Alibaba Cloud EBS afterwards, e.g., EBS ESSD, Local Storage Instance D3c [paper].

Publications

Book Chapters

2019 **Linux Open Source Storage: from Ceph to Container (Chinese Edition)**
In the Publishing House of Electronics Industry [book]

Conferences

2026 **SmoothAgent: Efficient Long-Horizon LLM-Based Agent Serving with Lookahead Context Engineering**
Zaifeng Pan, Qianxu Wang, Zhengding Hu, Chang Chen, Yue Guan, **Yanbo Zhou**, Steven Swanson, and Yufei Ding.
arXiv preprint arXiv:2607.00151 [paper]

2026 **Break on Through to the Other Side: Pooling Memory Elastically with RamRyder**
Yanbo Zhou, Erci Xu, Dongjoo Seo, Adam Manzanares, Steven Swanson.
USENIX Symposium on Operating Systems Design and Implementation (OSDI) [paper] [project]

2026 **Here, There and Everywhere: The Past, the Present and the Future of Local Storage in Cloud**
Leping Yang, **Yanbo Zhou**, Gong Zeng, Li Zhang, Saisai Zhang, Ruilin Wu, Chaoyang Sun, Shiyi Luo, Wenrui Li, Keqiang Niu, Xiaolu Zhang, Junping Wu, Jiaji Zhu, Jiesheng Wu, Mariusz Barczak, Wayne Gao, Ruiming Lu, Erci Xu, and Guangtao Xue
USENIX Conference on File and Storage Technologies (FAST) [paper]
Best Paper Award

2025 **Sleeping with One Eye Open: Fast, Sustainable Storage with Sandman**
Yanbo Zhou, Erci Xu, Anisa Su, Jim Harris, Adam Manzanares, Steven Swanson.
The Symposium on Operating Systems Principles (SOSP) [paper]

2024 **CSAL: the Next-Gen Local Disks for the Cloud**
Yanbo Zhou, Erci Xu, Li Zhang, Kapil Karkra, Mariusz Barczak, Wayne Gao, Wojciech Malikowski, Mateusz Kozlowski, Łukasz Łasek, Ruiming Lu, Feng Yang, Lilong Huang, Xiaolu Zhang, Wenrui Li, Jinhu Li, Keqiang Niu, Jiaji Zhu, Jiesheng Wu.
The European Conference on Computer Systems (EuroSys) [paper] [project]

2020 **Spool: Reliable Virtualized NVMe Storage Pool in Public Cloud Infrastructure.**
Shuai Xue, Shang Zhao, Quan Chen, Gang Deng, Zheng Liu, Jie Zhang, Zhuo Song, Tao Ma, Yong Yang, **Yanbo Zhou**, Keqiang Niu, Sijie Sun, Minyi Guo.
USENIX Annual Technical Conference (ATC) [paper]

2018 **Write-aware Data Allocation on Heterogeneous Memory Architecture with Minimum Cost.**
Yanbo Zhou, Shouzhen Gu, Lixia Zheng, Edwin H.-M. Sha, Qingfeng Zhuge.
The Conference on Embedded and Real-Time Computing System and Applications (RTCSA) [paper]

2018 **Accelerating I/Os in virtual machines on physical NVMe SSDs via user space vhost target.**
Ziye Yang, Changpeng Liu, **Yanbo Zhou**, Xiaodong Liu, Gang Cao
The Symposium on Cloud and Services Computing (SC2) [paper]

Tech Reports

2023 **A Media-Aware Cloud Storage Acceleration Layer (CSAL) Cache Solution with Intel Optane SSDs for Alibaba ECS Local Disk D3C Service.**
Yanbo Zhou, Li Zhang, Kapil Karkra, Wayne Gao, Chunhong Mao, Mariusz Barczak.
Intel White Paper [paper].

Talks

2026 **Break on Through to the Other Side: Pooling Memory Elastically with RamRyder**
OSDI'26
UCLA System Group
ACE Center for Evolvable Computing

2026 **Here, There and Everywhere: The Past, the Present and the Future of Local Storage in Cloud**
Meta's AI and Systems Co-Design Team
FAST'26 [slides] [talk]

2025 **Sleeping with One Eye Open: Fast, Sustainable Storage with Sandman**
ACE Center for Evolvable Computing
Meta's AI and Systems Co-Design Team [slides]

2024 **CSAL: the Next-Gen Local Disks for the Cloud**
EuroSys'24 [slides]

2023 **Best SPDK Practices: Lessons from Five Years of Storage Evolution in Alibaba Cloud**
SPDK PRC Forum [slides] [talk - chinese]

2022 **Removing QLC Write-Amplification through Intel Optane SSD with SPDK WSR**
SPDK PRC Forum [slides] [talk - chinese]

Teaching Experience

2025 **Introduction to Computer Architecture: A Software Perspective (UCSD CSE 142)**
Head Teaching Assistant
Head TA for an undergraduate computer architecture course with 90+ students; led four tutors and collaborated with instructor (Prof. Steven Swanson) to design assignments, quizzes, and exams. Held office hours to help students debug and reason through technical issues related to computer architecture.

Academic Service

Reviewer

- IEEE Computer Architecture Letters (CAL)

Artifact Evaluation Committee

- USENIX FAST'26, USENIX OSDI'26
- ACM ASPLOS'25, ACM EuroSys'25, ACM SOSP'25

Professional Skills

Hardware

- Memory: DRAM, Persistent Memory, CXL Memory (expert)
- Storage: CMR/SMR HDD, NVMe SSD, SLC/MLC/TLC/QLC NAND Flash Media (expert)
- Accelerator: Data Center DPU (expert)
- Network: RDMA NIC (some familiarity)

Software

- Storage Systems: Kernel Bypass SPDK, Storage Cache Design (expert)
- Virtualization: QEMU, Vhost-User (expert)
- Linux Kernel: filesystem, block layer, memory management (knowledgeable)
- Protocol: NVMe, NVMe-oF, Virtio (knowledgeable)
- Network Systems: Kernel Bypass DPDK (some familiarity)

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